

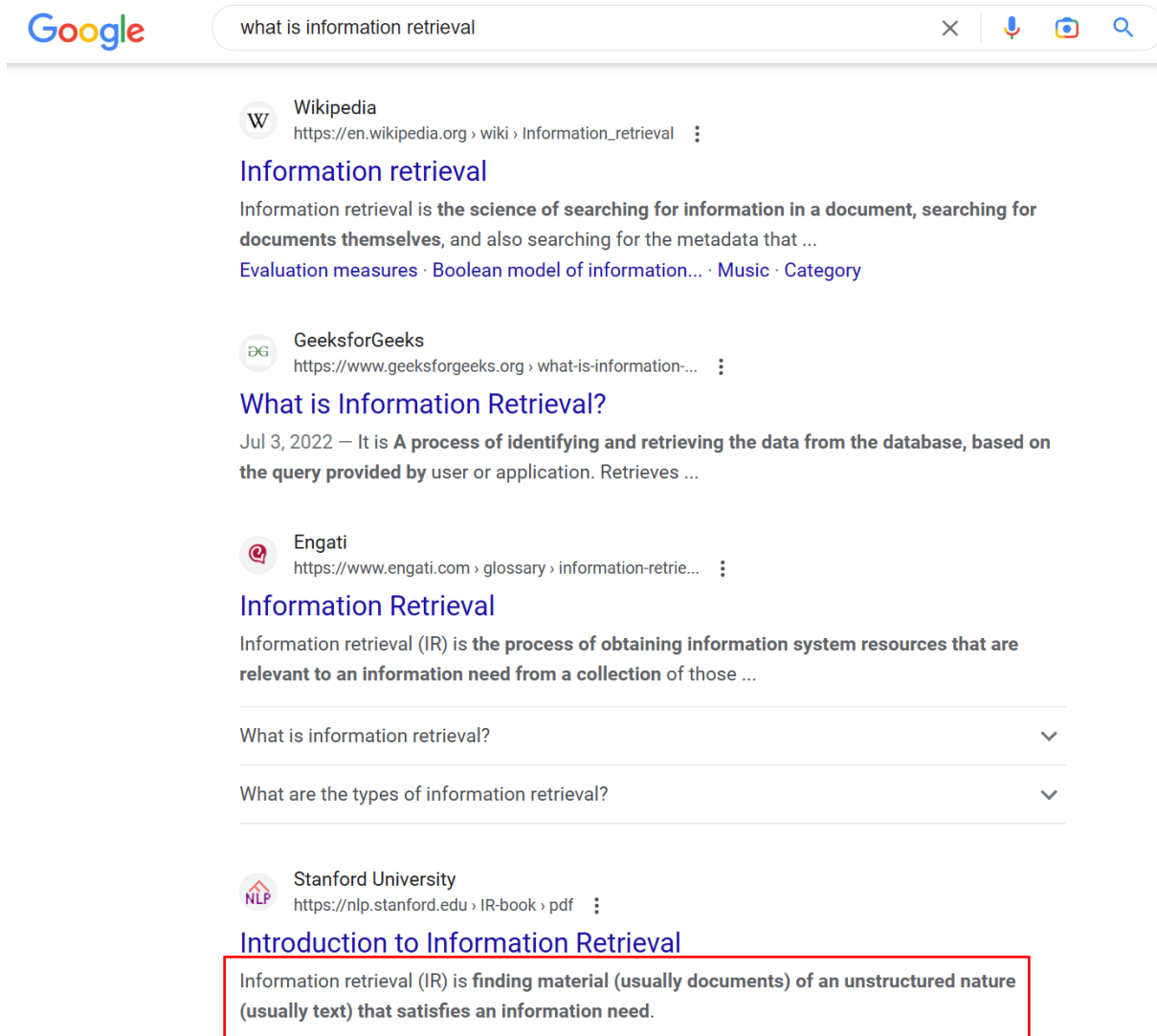
CS499: Information Retrieval

Lecture 2: Crawling and text processing

Huazheng Wang

EECS@OSU

Recap: What is information retrieval?



The screenshot shows a Google search for "what is information retrieval". The search bar is at the top with the Google logo on the left and search icons on the right. Below the search bar, there are three search results. The first result is from Wikipedia, titled "Information retrieval", with a snippet: "Information retrieval is the science of searching for information in a document, searching for documents themselves, and also searching for the metadata that ...". The second result is from GeeksforGeeks, titled "What is Information Retrieval?", with a snippet: "Jul 3, 2022 — It is A process of identifying and retrieving the data from the database, based on the query provided by user or application. Retrieves ...". The third result is from Engati, titled "Information Retrieval", with a snippet: "Information retrieval (IR) is the process of obtaining information system resources that are relevant to an information need from a collection of those ...". Below these results, there are two expandable sections: "What is information retrieval?" and "What are the types of information retrieval?". At the bottom, there is a result from Stanford University, titled "Introduction to Information Retrieval", with a snippet: "Information retrieval (IR) is finding material (usually documents) of an unstructured nature (usually text) that satisfies an information need." This snippet is highlighted with a red border.

Google

what is information retrieval

Wikipedia
https://en.wikipedia.org › wiki › Information_retrieval

Information retrieval

Information retrieval is the science of searching for information in a document, searching for documents themselves, and also searching for the metadata that ...

Evaluation measures · Boolean model of information... · Music · Category

GeeksforGeeks
https://www.geeksforgeeks.org › what-is-information-...

What is Information Retrieval?

Jul 3, 2022 — It is A process of identifying and retrieving the data from the database, based on the query provided by user or application. Retrieves ...

Engati
https://www.engati.com › glossary › information-retrie...

Information Retrieval

Information retrieval (IR) is the process of obtaining information system resources that are relevant to an information need from a collection of those ...

What is information retrieval?

What are the types of information retrieval?

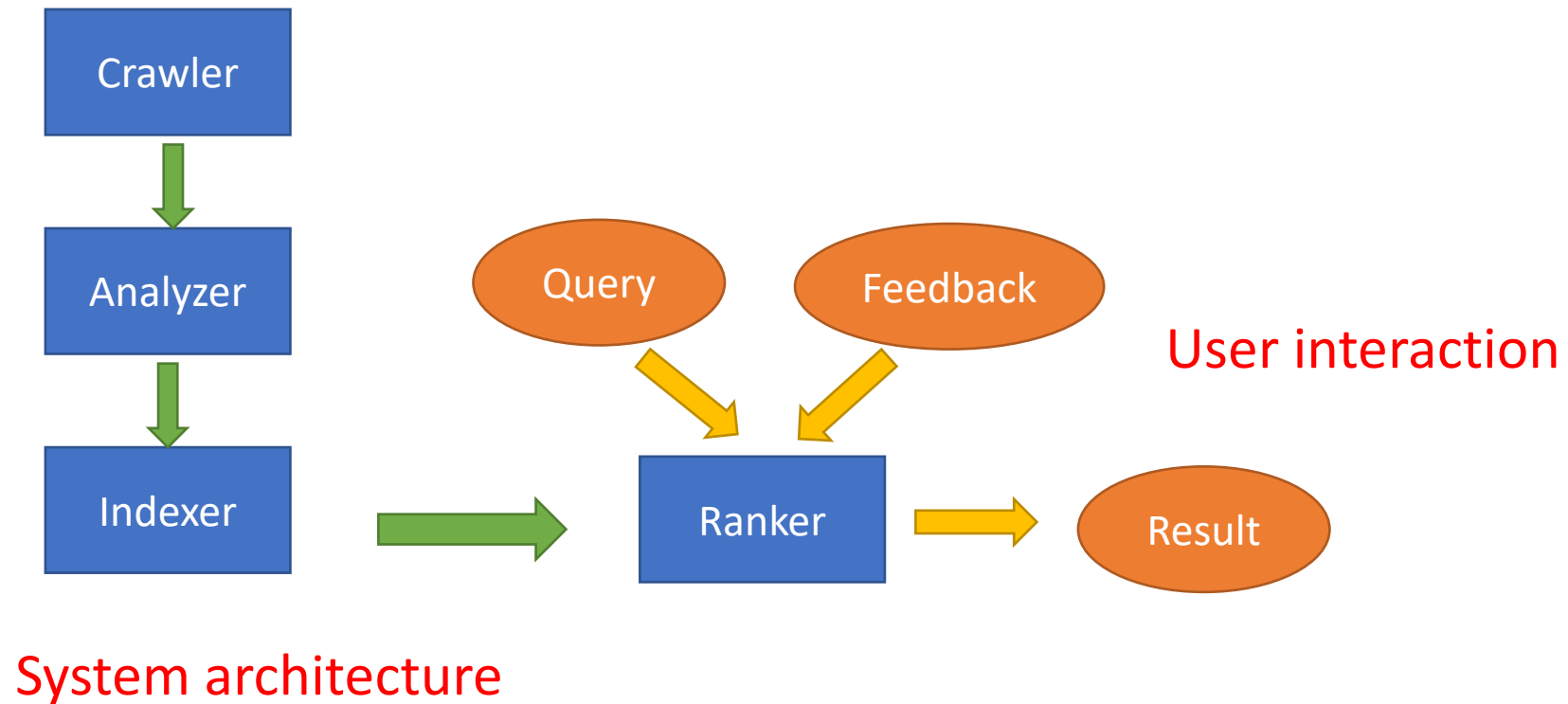
Stanford University
https://nlp.stanford.edu › IR-book › pdf

Introduction to Information Retrieval

Information retrieval (IR) is finding material (usually documents) of an unstructured nature (usually text) that satisfies an information need.

- Information Retrieval (IR) is **finding material** (usually documents) of an **unstructured** nature (usually text) that satisfies an **information need** from within large collections (usually stored on computers).

Recap: Search engine architecture



Web crawler

- A crawler is an automated program that systematically browses the World Wide Web, typically for the purpose of web indexing.
 - Also known as a spider, bot, or web crawler



What any crawler *must* do

- Be Robust: Be immune to spider traps and other malicious/non-malicious behavior from web servers
 - Spam pages, Spider traps – dynamically generated
 - Site mirrors and duplicate pages
- Be Polite: Respect implicit and explicit politeness considerations

Web crawler pseudo code

```
Def Crawler(entry_point) {  
    URL_list = [entry_point]  
    while (len(URL_list)>0) { Which page to visit next?  
        URL = URL_list.pop();  
        if (isVisited(URL) or !isLegal(URL) or !checkRobotsTxt(URL))  
            continue; Is it visited already?  
Or shall we visit it again?  
        HTML = URL.open(); Is the access granted?  
        for (anchor in HTML.listOfAnchors()) {  
            URL_list.append(anchor);  
        }  
        setVisited(URL);  
        insertToIndex(HTML);  
    }  
}
```

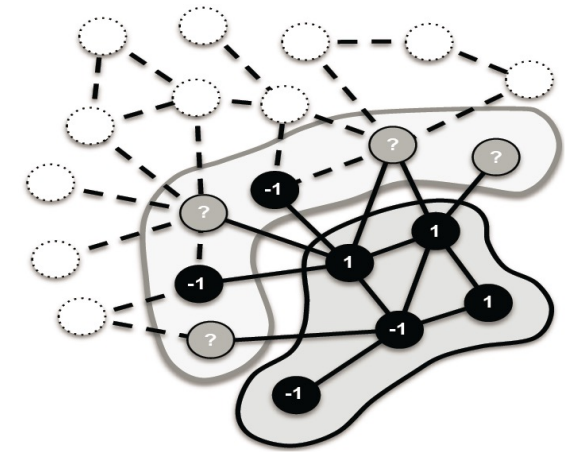
Visiting strategy

- Breadth first
 - Uniformly explore from the entry page
 - Memorize all nodes on the previous level
 - As shown in pseudo code
- Depth first
 - Explore the web by branch
 - Biased crawling given the web is not a tree structure
- Focused crawling
 - Prioritize the new links by predefined strategies

Focused crawling

- Prioritize the visiting sequence of the web
 - The size of Web is too large for a crawler (even Google) to completely cover
 - Not all documents are equally important
 - Emphasize more on the high-quality documents
 - Maximize weighted coverage

Focused crawling



- Prioritize by in-degree [Cho et al. WWW'98]
 - The page with the highest number of incoming hyperlinks from previously crawled pages is crawled next
- Prioritize by PageRank [Abiteboul et al. WWW'07, Cho and Uri VLDB'07]
 - Breadth-first in early stage, then compute/approximate PageRank periodically
 - More consistent with search relevance [Fetterly et al. SIGIR'09]
- Prioritize by topical relevance
 - In vertical search, only crawl relevant pages [De et al. WWW'94]
 - E.g., restaurant search engine should only crawl restaurant pages
 - Estimate the similarity to current page by anchor text or text near anchor [Hersovici et al. WWW'98]
 - User given taxonomy or topical classifier [Chakrabarti et al. WWW'98]

Avoid duplicate visits

- Given web is a graph rather than a tree, avoid loop in crawling is important
- How to check
 - trie or hash table
- What to check
 - URL: must be normalized, not necessarily can avoid all duplication
 - <http://dl.acm.org/event.cfm?id=RE160&CFID=516168213&CFTOKEN=99036335>
 - <http://dl.acm.org/event.cfm?id=RE160>
 - Page: minor change might cause misfire
 - Timestamp, data center ID change in HTML

Politeness policy

- Crawlers can retrieve data much quicker and in greater depth than human searchers
- Costs of using Web crawlers
 - Network resources
 - Server overload
- Robots exclusion protocol
 - robots.txt

Robots exclusion protocol

- Exclude specific directories:

User-agent: *

Disallow: /tmp/

Disallow: /cgi-bin/

Disallow: /users/paranoid/

- Exclude a specific robot:

User-agent: GoogleBot

Disallow: /

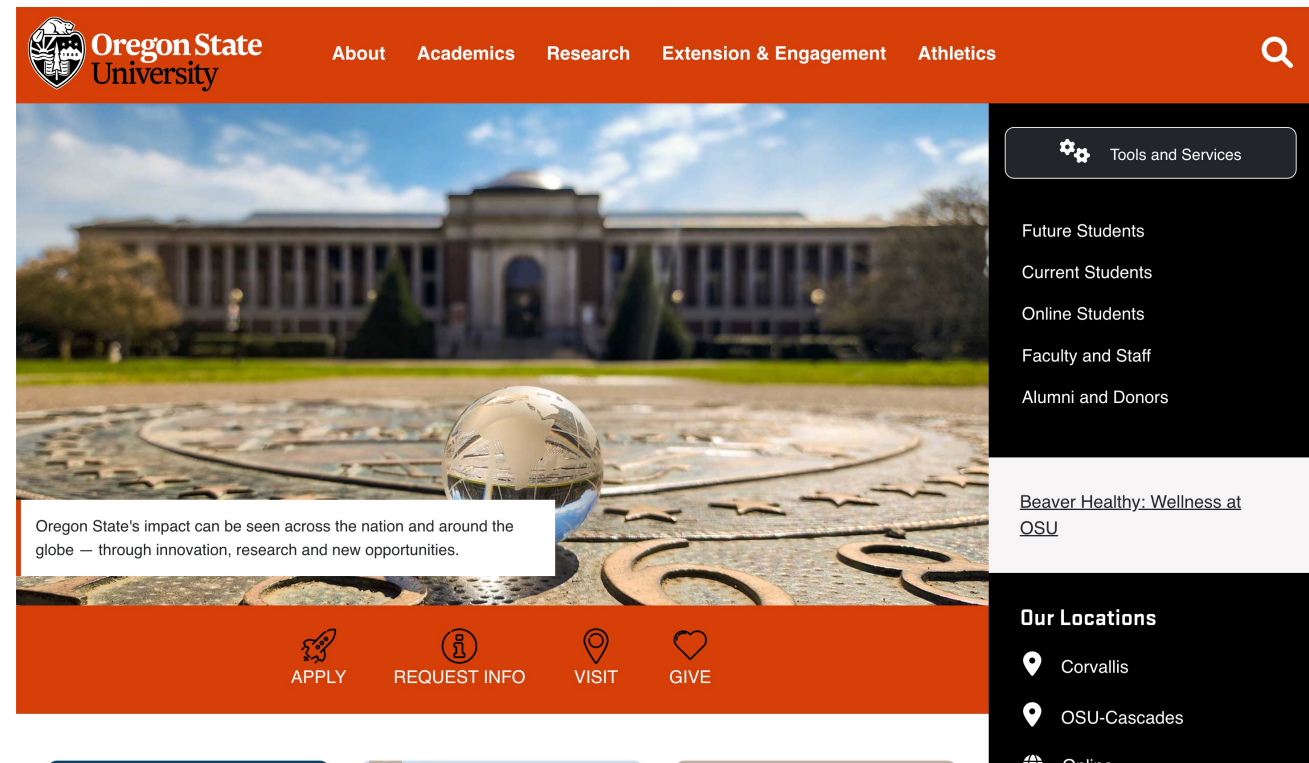
- Allow a specific robot:

User-agent: GoogleBot

Disallow:

User-agent: *

Disallow: /



Robots exclusion protocol

- Exclude specific directories:

```
User-agent: *
```

```
Disallow: /tmp/
```

```
Disallow: /cgi-bin/
```

```
Disallow: /users/paranoid/
```

- Exclude a specific robot:

```
User-agent: GoogleBot
```

```
Disallow: /
```

- Allow a specific robot:

```
User-agent: GoogleBot
```

```
Disallow:
```

```
User-agent: *
```

```
Disallow: /
```

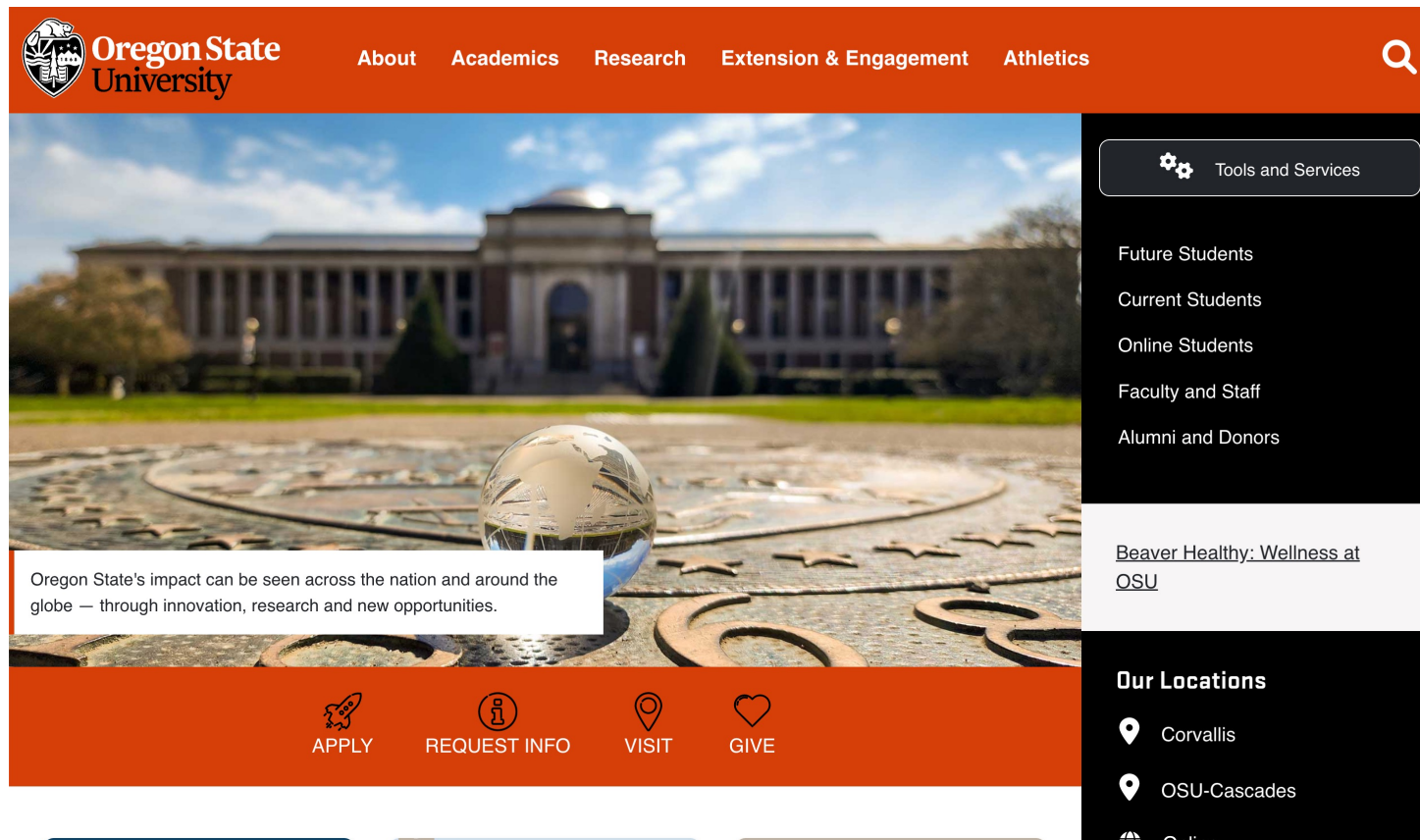
```
← ↻ 🏠 🔒 https://oregonstate.edu/robots.txt

#
# robots.txt
#
# This file is to prevent the crawling and indexing of certain parts
# of your site by web crawlers and spiders run by sites like Yahoo!
# and Google. By telling these "robots" where not to go on your site,
# you save bandwidth and server resources.
#
# This file will be ignored unless it is at the root of your host:
# Used:    http://example.com/robots.txt
# Ignored: http://example.com/site/robots.txt
#
# For more information about the robots.txt standard, see:
# http://www.robotstxt.org/robotstxt.html

User-agent: *
# CSS, JS, Images
Allow: /core/*.css$
Allow: /core/*.css?
Allow: /core/*.js$
Allow: /core/*.js?
Allow: /core/*.gif
Allow: /core/*.jpg
Allow: /core/*.jpeg
Allow: /core/*.png
Allow: /core/*.svg
Allow: /profiles/*.css$
Allow: /profiles/*.css?
Allow: /profiles/*.js$
Allow: /profiles/*.js?
Allow: /profiles/*.gif
Allow: /profiles/*.jpg
Allow: /profiles/*.jpeg
Allow: /profiles/*.png
Allow: /profiles/*.svg
# Directories
Disallow: /core/
Disallow: /profiles/
# Files
Disallow: /README.txt
Disallow: /web.config
# Paths (clean URLs)
Disallow: /admin/
Disallow: /comment/reply/
Disallow: /filter/tips
Disallow: /node/add/
Disallow: /search/
Disallow: /user/register
Disallow: /user/password
```

Analyze the web page

- What you want to crawl:



Analyze the web page

- Computer's view:

```
1 <!DOCTYPE html>
2 <html lang="en" dir="ltr" prefix="content: http://purl.org/rss/1.0/modules/content/ dc: http://purl.org/dc/terms/ foaf: http://xmlns.com/foaf/0.1/ og: http://ogp.
3   <head>
4     <meta charset="utf-8" /><script type="text/javascript">(window.NREUM||(NREUM={})).init={ajax:{deny_list:["bam.nr-data.net"]}};(window.NREUM||(NREUM={})).loader_c
5   <script async src="https://www.googletagmanager.com/gtag/js?id=G-OC3MB7K855"></script>
6   <script>window.dataLayer = window.dataLayer || [];function gtag(){dataLayer.push(arguments)};gtag("js", new Date());gtag("set", "developer_id.dMDhkMT", true);gtag("c
7   <link rel="canonical" href="https://oregonstate.edu/" />
8   <meta name="robots" content="index, follow" />
9   <link rel="shortlink" href="https://oregonstate.edu/" />
10  <link rel="image_src" href="https://oregonstate.edu/sites/default/files/paragraphs/linc-quad_0.jpg" />
11  <meta name="description" content="Oregon State University continues to advance its 150-year mission to provide access to a high-quality education as Oregon's la
12  <meta property="og:site_name" content="Oregon State University" />
13  <meta property="og:type" content="website" />
14  <meta property="og:url" content="https://oregonstate.edu/" />
15  <meta property="og:title" content="Oregon State University" />
16  <meta property="og:description" content="Oregon State University delivers exceptional, accessible education and problem-solving innovation as Oregon's largest a
17  <meta property="og:image" content="https://oregonstate.edu/sites/default/files/2021-03/linc-quad-1200x630.jpg" />
18  <meta property="og:image:width" content="1200" />
19  <meta property="og:image:height" content="630" />
20  <meta name="google-site-verification" content="EzfIeVj5D_Jkpdgh4JjLnaS6DFi762P2ijQnLgcU6t4" />
21  <meta name="twitter:card" content="summary_large_image" />
22  <meta name="twitter:site" content="@oregonstate" />
23  <meta name="twitter:description" content="Oregon State University delivers exceptional, accessible education and problem-solving innovation as Oregon's largest
24  <meta name="twitter:title" content="Oregon State University" />
25  <meta name="twitter:url" content="https://oregonstate.edu/" />
26  <meta name="twitter:image:height" content="1200" />
27  <meta name="twitter:image:width" content="630" />
28  <meta name="twitter:image" content="https://oregonstate.edu/sites/default/files/2021-03/linc-quad-1200x630.jpg" />
29  <meta name="Generator" content="Drupal 9 (https://www.drupal.org)" />
30  <meta name="MobileOptimized" content="width" />
31  <meta name="HandheldFriendly" content="true" />
32  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
33  <meta name="theme-color" media="(prefers-color-scheme: light)" content="#fff" />
34  <meta name="theme-color" media="(prefers-color-scheme: dark)" content="#000" />
35  <link rel="icon" href="/themes/osu/homepage/favicon.ico" type="image/vnd.microsoft.icon" />
36  <script src="/sites/default/files/google_tag/main_osu_container/google_tag.script.js?rsrc4f" defer></script>
37
38  <title>Oregon State University</title>
39  <link rel="stylesheet" media="all" href="/sites/default/files/css/css_OU7w3LS7kYFpWnOiiZteWTjS9-kvcOXrCtghqYldVow.css" />
40  <link rel="stylesheet" media="all" href="/sites/default/files/css/css_hN6WEVScBYcpkzIX-NHqJ_-VoCqZ4ClpJbXOIu79zC4.css" />
41  <link rel="stylesheet" media="all" href="//dlazclqln24ryf.cloudfront.net/155267/OregonStateBrandIcons/style-cf.css?lh540y" />
42  <link rel="stylesheet" media="all" href="//cdnjs.cloudflare.com/ajax/libs/font-awesome/6.1.1/css/all.min.css" />
43  <link rel="stylesheet" media="all" href="/sites/default/files/css/css_s_isNJebD0hx5FuRzRbK4BYPKmi3xHuo_od0TeK0KjM.css" />
44
45
46 </head>
47 <body>
48 <!-- Monsido Site Script -->
49 <script type="text/javascript">
50   window._monido = window._monido || {
51     token: "7BqhRyYP9fteuxY3jflurQ",
```

Parse HTML

- Generally difficult due to the free style of HTML

- Solutions

- Shallow parsing

- Remove all HTML tags
 - Only keep text between `<title></title>` and `<p></p>`

- Automatic wrapper generation [Crescenzi et al. VLDB'01]

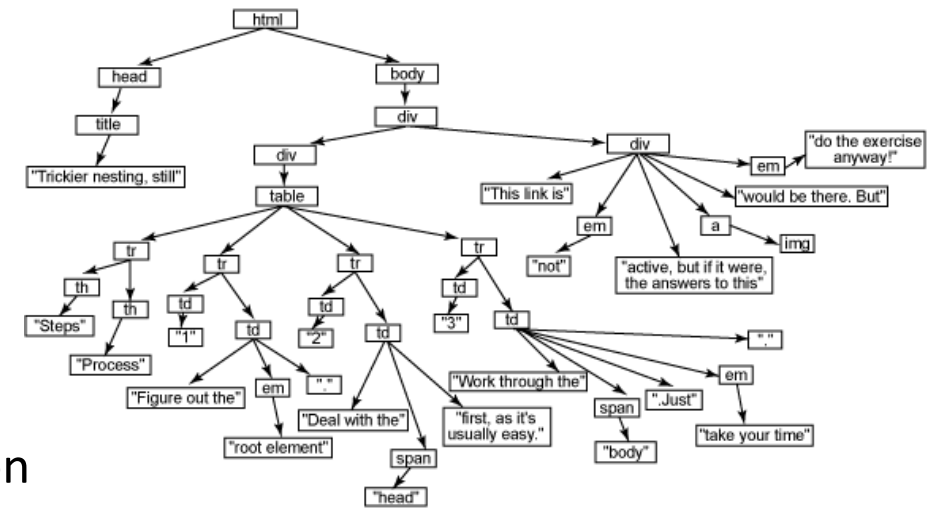
- Wrapper: regular expression for HTML tags' combination
 - Inductive reasoning from examples

- Visual parsing [Yang and Zhang DAR'01]

- Frequent pattern mining of visually similar HTML blocks

- Many popular packages

- Python: BeautifulSoup



How to represent a document/web page?

- Represent by a string?
 - No semantic meaning
- Represent by a list of sentences?
 - Sentence is just like a short document (recursive definition)
- Represent by a list of words?
 - Tokenize it first
 - Bag-of-Words representation!

Tokenization

- Break a stream of text into meaningful units
 - Tokens: words, phrases, symbols
 - Input: It's not straight-forward to perform so-called "tokenization."
 - Output(1): 'It's', 'not', 'straight-forward', 'to', 'perform', 'so-called', '"tokenization."'
 - Output(2): 'It', "'", 's', 'not', 'straight', '-', 'forward', 'to', 'perform', 'so', '-', 'called', '"', 'tokenization', '.', ' "'
- Definition depends on language, corpus, or even context



Tokenize the sentence: It's not straight-forward to perform so-called "tokenization."



Here's the tokenized sentence:

CSS

Copy code

```
['It', "'", 's', 'not', 'straight', '-', 'forward', 'to', 'perform', 'so', '-', 'called', '"', 'tokenization', '.', ' "'
```

In this tokenization, each word is treated as a separate token, including hyphenated words and punctuation. The apostrophe in "It's" has been split into two separate tokens, "It" and "s". The double quotes around "tokenization" have also been treated as separate tokens.

Tokenization

- Solutions
 - Regular expression
 - `[\w]+`: so-called -> 'so', 'called'
 - `[\S]+`: It's -> 'It's' instead of 'It', 's'
 - Statistical methods
 - Explore rich features to decide where is the boundary of a word
 - Apache OpenNLP (<http://opennlp.apache.org/>)
 - Stanford NLP Parser (<http://nlp.stanford.edu/software/lex-parser.shtml>)
 - NLTK ([NLTK :: Natural Language Toolkit](#))
 - spaCy ([spaCy · Industrial-strength Natural Language Processing in Python](#))

Text indexing

- Bag-of-Words representation

Terms	Documents													
	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14
abnormalities	0	0	0	0	0	0	0	1	0	1	0	0	0	0
age	1	0	0	0	0	0	0	0	0	0	0	1	0	0
behavior	0	0	0	0	1	1	0	0	0	0	0	0	0	0
blood	0	0	0	0	0	0	0	1	0	0	1	0	0	0
close	0	0	0	0	0	0	1	0	0	0	1	0	0	0
culture	1	1	0	0	0	0	0	1	1	0	0	0	0	0
depressed	1	0	1	1	1	0	0	0	0	0	0	0	0	0
discharge	1	1	0	0	0	1	0	0	0	0	0	0	0	0
disease	0	0	0	0	0	0	0	0	1	0	1	0	0	0
fast	0	0	0	0	0	0	0	0	0	1	0	1	1	1
generation	0	0	0	0	0	0	0	0	1	0	0	0	1	0
oestrogen	0	0	1	1	0	0	0	0	0	0	0	0	0	0
patients	1	1	0	1	0	0	0	1	0	0	0	0	0	0
pressure	0	0	0	0	0	0	0	0	0	0	1	0	0	1
rats	0	0	0	0	0	0	0	0	0	0	0	0	1	1
respect	0	0	0	0	0	0	0	1	0	0	0	1	0	0
rise	0	0	0	1	0	0	0	0	0	0	0	0	0	1
study	1	0	1	0	0	0	0	0	1	0	0	0	0	0

- Doc1: Information retrieval is helpful for everyone.
- Doc2: Helpful information is retrieved for you.

	information	retrieval	retrieved	is	helpful	for	you	everyone
Doc1	1	1	0	1	1	1	0	1
Doc2	1	0	1	1	1	1	1	0

Word-document adjacency matrix



Text indexing

- Bag-of-Words representation
 - Assumption: word is independent from each other
 - Pros: simple
 - Cons: grammar and order are missing
 - *(Used to be) The most frequently used document representation*
 - *Image, speech, gene sequence*
- Improved Bag-of-Words representation
 - N-grams: a contiguous sequence of n items from a given sequence of text
 - E.g., Information retrieval is helpful for everyone
 - Bigrams: 'information_retrieval', 'retrieval_is', 'is_helpful', 'helpful_for', 'for_everyone'
 - Pros: capture local dependency and order
 - Cons: purely statistical view, increase vocabulary size $O(V^N)$

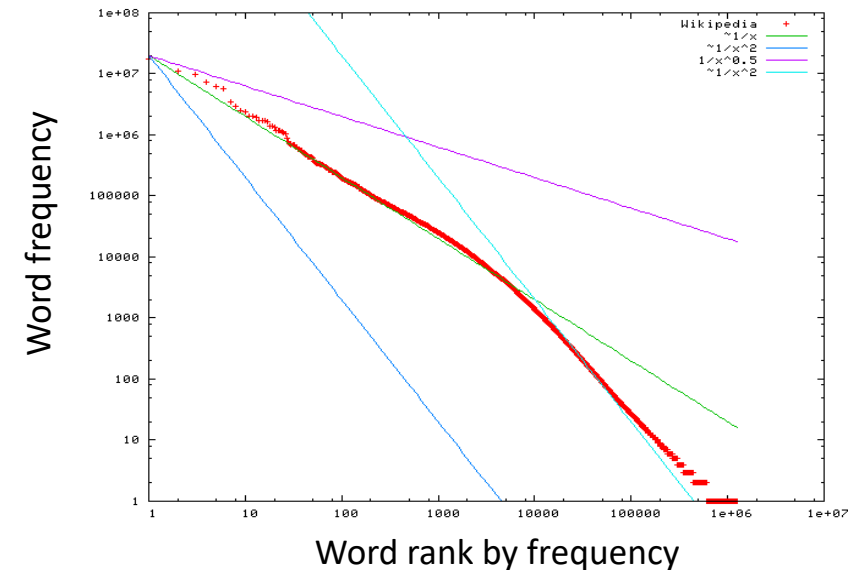
Text indexing

- Index document with all the occurring word
 - Pros
 - Preserve all information in the text (hopefully)
 - Fully automatic
 - Cons
 - Vocabulary gap: cars v.s., car
 - Large storage: e.g., in N-grams $O(V^N)$
 - Solution
 - Construct controlled vocabulary

Statistical property of language

- Zipf's law
 - Frequency of any word is inversely proportional to its rank in the frequency table
 - Formally
 - $f(k; s, N) = \frac{1/k^s}{\sum_{n=1}^N 1/n^s}$ where k is rank of the word; N is the vocabulary size; s is a language-specific parameter

*In the Brown Corpus of American English text, the word "the" is the most frequently occurring word, and by itself accounts for nearly **7%** of all word occurrences; the second-place word "of" accounts for slightly over **3.5%** of words.*



A plot of word frequency in Wikipedia (Nov 27, 2006)

Message of Zipf's law

- Head words may take large portion of occurrence, but they are semantically meaningless
 - E.g., the, a, an, we, do, to
- Tail words take major portion of vocabulary, but they rarely occur in documents
 - E.g., dextrosinistral
- The rest is most representative
 - To be included in the controlled vocabulary

Stopwords

- Useless words for query/document analysis
 - Not all words are informative
 - Remove such words to reduce vocabulary size
 - No universal definition
 - Risk: break the original meaning and structure of text
 - E.g., this is not a good option -> option
to be or not to be -> null

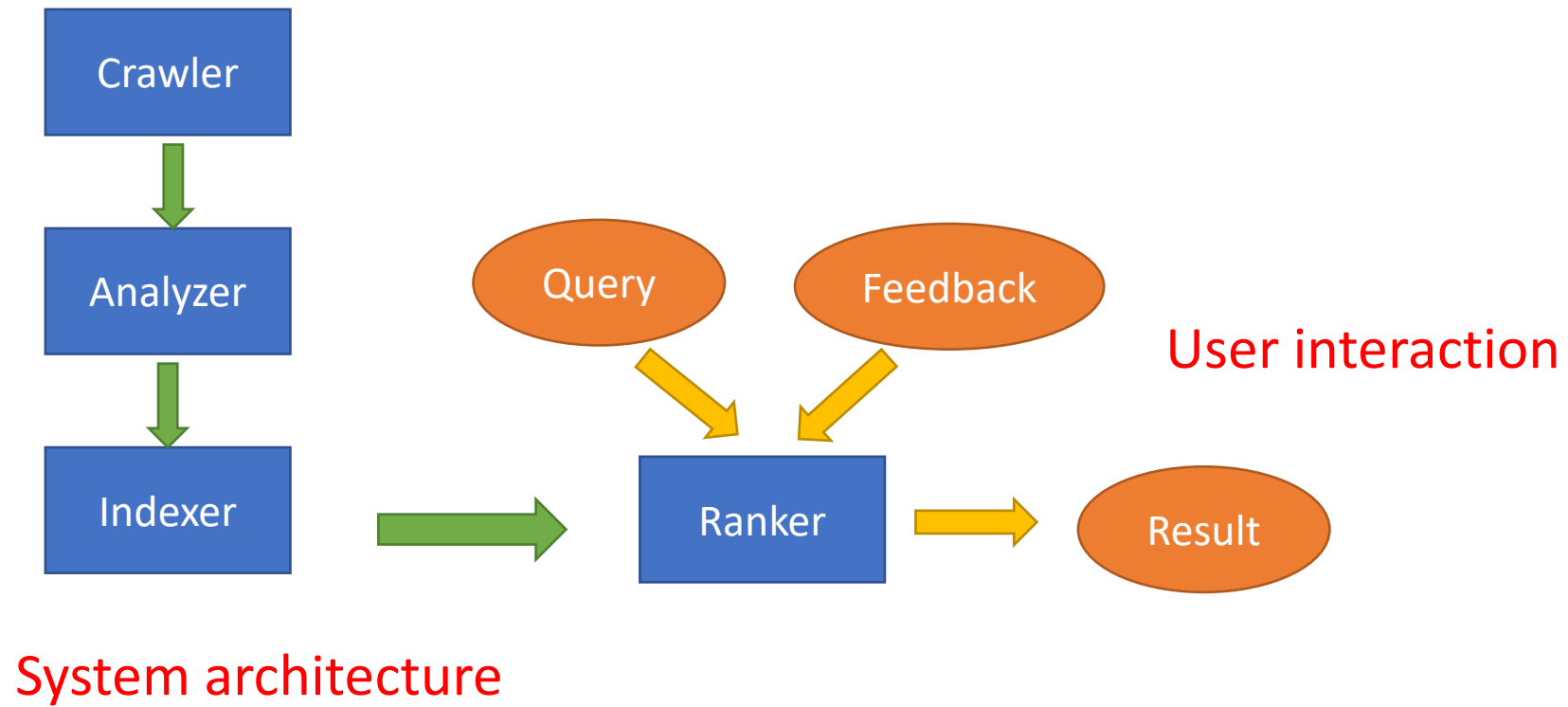
Normalization

- Convert different forms of a word to normalized form in the vocabulary
 - U.S.A -> USA, St. Louis -> Saint Louis
- Solution
 - Rule-based
 - Delete periods and hyphens
 - All in lower case
 - Dictionary-based
 - Construct equivalent class
 - Car -> “automobile, vehicle”
 - Mobile phone -> “cellphone”

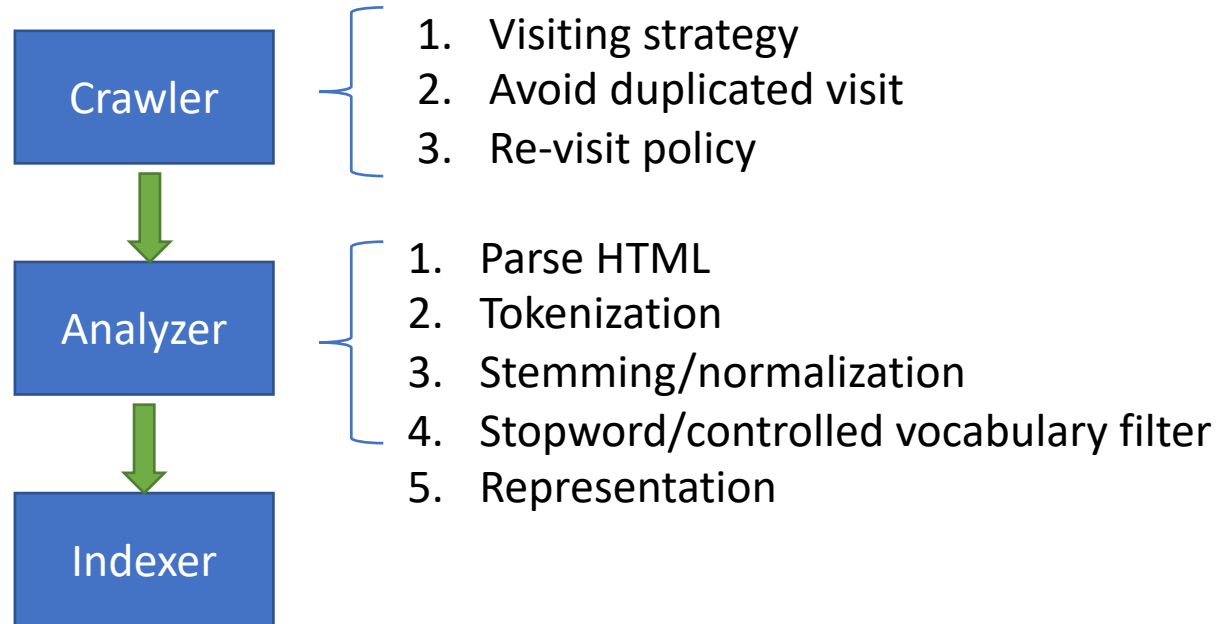
Stemming

- Reduce inflected or derived words to their root form
 - Plurals, adverbs, inflected word forms
 - E.g., ladies -> ladi, referring -> refer, forgotten -> forget
 - Bridge the vocabulary gap
 - Risk: lose precise meaning of the word
 - E.g., lay -> lie (a false statement? or be in a horizontal position?)
 - Solutions (for English)
 - Porter stemmer: pattern of vowel-consonant sequence
 - Krovetz stemmer: morphological rules

Search engine architecture



Search engine architecture



Text indexing

- In modern search engine
 - **No** stemming or stopword removal, since computation and storage are no longer the major concerns
 - Preserve more semantics. Recall the query *“to be or not to be”*
 - More advanced NLP techniques are applied
 - Named entity recognition
 - E.g., people, location and organization
 - Dependency parsing
 - Word/Document embeddings

Summary

- Web crawler basics
- Zipf's law
- Procedures for text indexing
- Bag-of-Words document representation

Readings

- [MRS] Ch 2, Ch20.